(C1) *Identify the decision alternatives, the states of nature, the uncertain event, and the payoff components.* Then **(C2)** *construct the corresponding payoff table.*

Context Descriptions

1. A school event company must choose between a compact snack stand and a full snack stand before attendance is known. Attendance may be high or low. Analysts have already converted each possible consequence into direct profit measured in thousand USD. The compact stand produces a profit of 18 thousand USD under high attendance and 11 thousand USD under low attendance. The full stand produces a profit of 27 thousand USD under high attendance and 6 thousand USD under low attendance.
2. A weekend pop-up business must choose between an indoor booth and a mobile stall before market demand is known. Demand may be strong or weak. The indoor booth contributes a fixed 550 thousand JPY to profit, while the mobile stall contributes a fixed 420 thousand JPY. Strong demand adds a fixed market bonus of 180 thousand JPY to either alternative. Weak demand instead reduces profit by a fixed 90 thousand JPY because of permit and cleanup expenses.
3. A local delivery firm must choose between a bicycle fleet and an electric-van fleet before traffic conditions are known. Traffic may be clear or congested. The bicycle fleet earns 4 thousand CNY per completed delivery batch and is expected to complete 14 delivery batches in clear traffic and 10 in congested traffic. The electric-van fleet earns 6 thousand CNY per completed delivery batch and is expected to complete 12 delivery batches in clear traffic and 8 in congested traffic. Clear traffic also adds a fixed scheduling bonus of 5 thousand CNY to either fleet, while congestion creates a fixed loss of 4 thousand CNY.
4. A city-tour operator must choose between a standard tour and a premium tour before tourism demand is known. Demand may be high or low. The standard tour contributes 2 thousand EUR per group booking and also provides a fixed contract contribution of 6 thousand EUR. The premium tour contributes 4 thousand EUR per group booking and provides a fixed contract contribution of 10 thousand EUR. High tourism demand adds another 3 thousand EUR per group booking to either tour, while low demand adds 1 thousand EUR per group booking. The standard tour is expected to complete 16 group bookings under high demand and 10 under low demand; the premium tour is expected to complete 12 group bookings under high demand and 7 under low demand.
5. A solar-installation company must choose between a standard installation crew and an expanded installation crew before customer orders are known. Orders may be strong, regular, or weak. The standard crew contributes 3 thousand GBP per completed installation and provides a fixed contract contribution of 8 thousand GBP. The expanded crew contributes 4 thousand GBP per completed installation and provides a fixed contract contribution of 5 thousand GBP. Strong orders add 2 thousand GBP per completed installation and a fixed scheduling bonus of 6 thousand GBP. Regular orders add 1 thousand GBP per completed installation and no fixed adjustment. Weak orders reduce the contribution by 1 thousand GBP per completed installation and create a fixed idle-time loss of 4 thousand GBP. The standard crew is expected to complete 10 installations under strong orders, 7 under regular orders, and 4 under weak orders. The expanded crew is expected to complete 14, 9, and 5 installations under the same respective states.
6. A coffee exporter must choose among contracted storage, a rented warehouse, and an automated distribution hub before export demand and shipping conditions are known. Conditions may be favorable, normal, or adverse. Contracted storage contributes 4 million COP per export lot handled but requires a fixed storage commitment of 12 million COP. The rented warehouse contributes 5 million COP per export lot but requires a fixed commitment of 20 million COP. The automated hub contributes 6 million COP per export lot but requires a fixed commitment of 32 million COP. Favorable conditions add 2 million COP per export lot and a fixed logistics bonus of 8 million COP. Normal conditions make no additional per-lot or fixed adjustment. Adverse conditions reduce the contribution by 2 million COP per export lot and create a fixed disruption loss of 10 million COP. Contracted storage is expected to handle 18 export lots under favorable conditions, 13 under normal conditions, and 8 under adverse conditions. The rented warehouse is expected to handle 22, 15, and 9 export lots, while the automated hub is expected to handle 26, 17, and 10 export lots under the same respective states.

1 Solutions: Extraction

Problems:

1.1. School Event Snack Stand (C1–C2)

1.2. Weekend Pop-Up Business (C1–C2)

1.3. Local Delivery Fleet (C1–C2)

1.4. City-Tour Format (C1–C2)

1.5. Solar-Installation Crew (C1–C2)

1.6. Coffee-Export Storage Strategy (C1–C2)

1.1 Exercise 1: School event snack stand

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Compact stand	A_2: Full stand
States of nature	S_1: High attendance	S_2: Low attendance
Uncertain event	Attendance level after the stand is selected	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. Monetary measure: Profit in thousand USD for each stand–attendance pair	



C2 - Building the payoff table.

Relevant values. The context supplies a direct profit for every stand–attendance pair.

Abstract component	Context component	Math symbol	Corresponding value
Direct payoff	Profit stated for each stand under each attendance level	$P_{ay}(A_i, S_j)$	Compact–high: 18 thousand USD; Compact–low: 11 thousand USD; Full–high: 27 thousand USD; Full–low: 6 thousand USD

Payoff equation. Because these are direct-payoff data, $P_{ay}(A_i, S_j)$ is read directly from the context.

Calculations.

Stand	High attendance	Low attendance
Compact stand	Direct context value: 18	Direct context value: 11
Full stand	Direct context value: 27	Direct context value: 6

Final payoff table (profit in thousand USD).

Alternative	High attendance (S_1)	Low attendance (S_2)
Compact stand (A_1)	18	11
Full stand (A_2)	27	6

1.2 Exercise 2: Weekend pop-up business

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1**:** Indoor booth	A_2**:** Mobile stall
States of nature	S_1**:** Strong demand	S_2**:** Weak demand
Uncertain event	Market demand after the selling format is selected	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand JPY for each format–demand pair.	



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Indoor booth; A_2: Mobile stall	Alternative labels, in row order.
S_j	S_1: Strong demand; S_2: Weak demand	State labels, in column order; revealed after the choice.
$f^A(A_i)$	Indoor booth: 550 thousand JPY; Mobile stall: 420 thousand JPY	Fixed contribution to profit from the chosen selling format.
$f^S(S_j)$	Strong demand: 180 thousand JPY; Weak demand: -90 thousand JPY	Market bonus under strong demand or permit-and-cleanup loss under weak demand.

Payoff equation.

$$P_{ay}(A_i, S_j) = f^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
	$f^A(A_1)$	$f^A(A_1)$
A_1	$+ f^S(S_1)$	$+ f^S(S_2)$
	$= P_{ay}(A_1, S_1)$	$= P_{ay}(A_1, S_2)$
	$f^A(A_2)$	$f^A(A_2)$
A_2	$+ f^S(S_1)$	$+ f^S(S_2)$
	$= P_{ay}(A_2, S_1)$	$= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
	550	550
A_1	$+180$	-90
	$=730$	$=460$
	420	420
A_2	$+180$	-90
	$=600$	$=330$

Negative rates reduce the per-unit contribution; negative fixed amounts represent losses or commitments deducted once. A negative final payoff is a shortfall.

Final payoff table (profit in thousand JPY).

Alternative	Strong demand (S_1)	Weak demand (S_2)
Indoor booth (A_1)	730	460
Mobile stall (A_2)	600	330

1.3 Exercise 3: Local delivery fleet

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Bicycle fleet	A_2: Electric-van fleet
States of nature	S_1: Clear traffic	S_2: Congested traffic
Uncertain event	Traffic conditions after the fleet is selected	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand CNY for each fleet–traffic pair.	



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Bicycle fleet; A_2: Electric-van fleet	Alternative labels, in row order.
S_j	S_1: Clear traffic; S_2: Congested traffic	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Bicycle–clear: 14 batches; Bicycle–congested: 10 batches; Electric van–clear: 12 batches; Electric van–congested: 8 batches	Expected completed delivery batches for each fleet under each traffic condition.
$v^A(A_i)$	Bicycle fleet: 4 thousand CNY/batch; Electric-van fleet: 6 thousand CNY/batch	Earnings per completed delivery batch for the chosen fleet.
$f^S(S_j)$	Clear traffic: 5 thousand CNY; Congested traffic: -4 thousand CNY	Scheduling bonus in clear traffic or congestion loss.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1, S_1) \times v^A(A_1)$ $+ f^S(S_1)$ $= P_{ay}(A_1, S_1)$	$q(A_1, S_2) \times v^A(A_1)$ $+ f^S(S_2)$ $= P_{ay}(A_1, S_2)$
A_2	$q(A_2, S_1) \times v^A(A_2)$ $+ f^S(S_1)$ $= P_{ay}(A_2, S_1)$	$q(A_2, S_2) \times v^A(A_2)$ $+ f^S(S_2)$ $= P_{ay}(A_2, S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	14×4 $+5$ $=61$	10×4 -4 $=36$
A_2	12×6 $+5$ $=77$	8×6 -4 $=44$

Negative rates reduce the per-unit contribution; negative fixed amounts represent losses or commitments deducted once. A negative final payoff is a shortfall.

Final payoff table (profit in thousand CNY).

Alternative	Clear traffic (S_1)	Congested traffic (S_2)
Bicycle fleet (A_1)	61	36
Electric-van fleet (A_2)	77	44

1.4 Exercise 4: City-tour format

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	A_1: Standard tour	A_2: Premium tour
States of nature	S_1: High tourism demand	S_2: Low tourism demand
Uncertain event	Tourism demand after the tour format is selected	
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand EUR for each tour–demand pair.	



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Standard tour; A_2: Premium tour	Alternative labels, in row order.
S_j	S_1: High tourism demand; S_2: Low tourism demand	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Standard–high: 16 bookings; Standard–low: 10 bookings; Premium–high: 12 bookings; Premium–low: 7 bookings	Expected group bookings for each tour under each tourism-demand level.
$v^A(A_i)$	Standard tour: 2 thousand EUR/booking; Premium tour: 4 thousand EUR/booking	Contribution per group booking from the chosen tour.
$v^S(S_j)$	High demand: 3 thousand EUR/booking; Low demand: 1 thousand EUR/booking	Additional contribution per group booking from tourism demand.
$f^A(A_i)$	Standard tour: 6 thousand EUR; Premium tour: 10 thousand EUR	Fixed contract contribution from the chosen tour.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2
A_1	$q(A_1,S_1)\times$ $[v^A(A_1)+v^S(S_1)]$ $+f^A(A_1)$ $=P_{ay}(A_1,S_1)$	$q(A_1,S_2)\times$ $[v^A(A_1)+v^S(S_2)]$ $+f^A(A_1)$ $=P_{ay}(A_1,S_2)$
A_2	$q(A_2,S_1)\times$ $[v^A(A_2)+v^S(S_1)]$ $+f^A(A_2)$ $=P_{ay}(A_2,S_1)$	$q(A_2,S_2)\times$ $[v^A(A_2)+v^S(S_2)]$ $+f^A(A_2)$ $=P_{ay}(A_2,S_2)$

Numerical calculations.

Alternative	S_1	S_2
A_1	$16\times$ $[2+3]$ $+6$ $=86$	$10\times$ $[2+1]$ $+6$ $=36$
A_2	$12\times$ $[4+3]$ $+10$ $=94$	$7\times$ $[4+1]$ $+10$ $=45$

Final payoff table (profit in thousand EUR).

Alternative	High tourism demand (S_1)	Low tourism demand (S_2)
Standard tour (A_1)	86	36
Premium tour (A_2)	94	45

1.5 Exercise 5: Solar-installation crew

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Standard installation crew	A_2: Expanded installation crew	
States of nature	S_1: Strong orders	S_2: Regular orders	S_3: Weak orders
Uncertain event	Customer-order level after the crew size is selected		
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in thousand GBP for each crew–order pair.		



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Standard installation crew; A_2: Expanded installation crew	Alternative labels, in row order.
S_j	S_1: Strong orders; S_2: Regular orders; S_3: Weak orders	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Standard–strong: 10; Standard–regular: 7; Standard–weak: 4 installations; Expanded–strong: 14; Expanded–regular: 9; Expanded–weak: 5 installations	Expected completed installations for each crew under each order level.
$v^A(A_i)$	Standard crew: 3 thousand GBP/installation; Expanded crew: 4 thousand GBP/installation	Contribution per completed installation from the chosen crew.
$v^S(S_j)$	Strong: 2 thousand GBP/installation; Regular: 1 thousand GBP/installation; Weak: -1 thousand GBP/installation	Per-installation addition or reduction from the order level.
$f^A(A_i)$	Standard crew: 8 thousand GBP; Expanded crew: 5 thousand GBP	Fixed contract contribution from the chosen crew.
$f^S(S_j)$	Strong: 6 thousand GBP; Regular: 0 thousand GBP; Weak: -4 thousand GBP	Scheduling bonus, no regular-order adjustment, or idle-time loss.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2	S_3
	$q(A_1, S_1)\times$	$q(A_1, S_2)\times$	$q(A_1, S_3)\times$
	$[v^A(A_1)+v^S(S_1)]$	$[v^A(A_1)+v^S(S_2)]$	$[v^A(A_1)+v^S(S_3)]$
A_1	$+f^A(A_1)$	$+f^A(A_1)$	$+f^A(A_1)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_1, S_1)$	$=P_{ay}(A_1, S_2)$	$=P_{ay}(A_1, S_3)$
	$q(A_2, S_1)\times$	$q(A_2, S_2)\times$	$q(A_2, S_3)\times$
	$[v^A(A_2)+v^S(S_1)]$	$[v^A(A_2)+v^S(S_2)]$	$[v^A(A_2)+v^S(S_3)]$
A_2	$+f^A(A_2)$	$+f^A(A_2)$	$+f^A(A_2)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_2, S_1)$	$=P_{ay}(A_2, S_2)$	$=P_{ay}(A_2, S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
	$10\times$	$7\times$	$4\times$
	$[3+2]$	$[3+1]$	$[3-1]$
A_1	$+8$	$+8$	$+8$
	$+6$	$+0$	-4
	$=64$	$=36$	$=12$
	$14\times$	$9\times$	$5\times$
	$[4+2]$	$[4+1]$	$[4-1]$
A_2	$+5$	$+5$	$+5$
	$+6$	$+0$	-4
	$=95$	$=50$	$=16$

Negative rates reduce the per-unit contribution; negative fixed amounts represent losses or commitments deducted once. A negative final payoff is a shortfall.

Final payoff table (profit in thousand GBP).

Alternative	Strong orders (S_1)	Regular orders (S_2)	Weak orders (S_3)
Standard installation crew (A_1)	64	36	12
Expanded installation crew (A_2)	95	50	16

1.6 Exercise 6: Coffee-export storage strategy

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	A_1: Contracted storage	A_2: Rented warehouse	A_3: Automated distribution hub
States of nature	S_1: Favorable conditions	S_2: Normal conditions	S_3: Adverse conditions
Uncertain event	Export-demand and shipping conditions after the storage strategy is selected		
Consequences	Operating the selected activity under the realised conditions leaves funds available or a shortfall to cover. The monetary measure is profit in million COP for each storage–condition pair.		



C2 - Building the payoff table.

Symbols, values, and meanings.

Symbol	Value (including units)	Meaning in this problem
A_i	A_1: Contracted storage; A_2: Rented warehouse; A_3: Automated distribution hub	Alternative labels, in row order.
S_j	S_1: Favorable conditions; S_2: Normal conditions; S_3: Adverse conditions	State labels, in column order; revealed after the choice.
$q(A_i, S_j)$	Contracted–favorable: 18; normal: 13; adverse: 8 lots; Rented–favorable: 22; normal: 15; adverse: 9 lots; Automated–favorable: 26; normal: 17; adverse: 10 lots	Expected export lots handled by each storage strategy under each condition.
$v^A(A_i)$	Contracted: 4 million COP/lot; Rented: 5 million COP/lot; Automated: 6 million COP/lot	Contribution per export lot from the chosen storage strategy.
$v^S(S_j)$	Favorable: 2 million COP/lot; Normal: 0 million COP/lot; Adverse: -2 million COP/lot	Per-lot addition, no adjustment, or reduction from conditions.
$f^A(A_i)$	Contracted: -12 million COP; Rented: -20 million COP; Automated: -32 million COP	Fixed storage commitment required by the chosen strategy.
$f^S(S_j)$	Favorable: 8 million COP; Normal: 0 million COP; Adverse: -10 million COP	Logistics bonus, no adjustment, or disruption loss from conditions.

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i) + f^S(S_j).$$

Additive payoff

Symbolic calculations.

Alternative	S_1	S_2	S_3
	$q(A_1,S_1)\times$	$q(A_1,S_2)\times$	$q(A_1,S_3)\times$
	$[v^A(A_1)+v^S(S_1)]$	$[v^A(A_1)+v^S(S_2)]$	$[v^A(A_1)+v^S(S_3)]$
A_1	$+f^A(A_1)$	$+f^A(A_1)$	$+f^A(A_1)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_1,S_1)$	$=P_{ay}(A_1,S_2)$	$=P_{ay}(A_1,S_3)$
	$q(A_2,S_1)\times$	$q(A_2,S_2)\times$	$q(A_2,S_3)\times$
	$[v^A(A_2)+v^S(S_1)]$	$[v^A(A_2)+v^S(S_2)]$	$[v^A(A_2)+v^S(S_3)]$
A_2	$+f^A(A_2)$	$+f^A(A_2)$	$+f^A(A_2)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_2,S_1)$	$=P_{ay}(A_2,S_2)$	$=P_{ay}(A_2,S_3)$
	$q(A_3,S_1)\times$	$q(A_3,S_2)\times$	$q(A_3,S_3)\times$
	$[v^A(A_3)+v^S(S_1)]$	$[v^A(A_3)+v^S(S_2)]$	$[v^A(A_3)+v^S(S_3)]$
A_3	$+f^A(A_3)$	$+f^A(A_3)$	$+f^A(A_3)$
	$+f^S(S_1)$	$+f^S(S_2)$	$+f^S(S_3)$
	$=P_{ay}(A_3,S_1)$	$=P_{ay}(A_3,S_2)$	$=P_{ay}(A_3,S_3)$

Numerical calculations.

Alternative	S_1	S_2	S_3
	$18\times$	$13\times$	$8\times$
	$[4+2]$	$[4+0]$	$[4-2]$
A_1	-12	-12	-12
	$+8$	$+0$	-10
	$=104$	$=40$	$=-6$
	$22\times$	$15\times$	$9\times$
	$[5+2]$	$[5+0]$	$[5-2]$
A_2	-20	-20	-20
	$+8$	$+0$	-10
	$=142$	$=55$	$=-3$
	$26\times$	$17\times$	$10\times$
	$[6+2]$	$[6+0]$	$[6-2]$
A_3	-32	-32	-32
	$+8$	$+0$	-10
	$=184$	$=70$	$=-2$

Negative rates reduce the per-unit contribution; negative fixed amounts represent losses or commitments deducted once. A negative final payoff is a shortfall.

Final payoff table (profit in million COP).

Alternative	Favorable (S_1)	Normal (S_2)	Adverse (S_3)
Contracted storage (A_1)	104	40	-6
Rented warehouse (A_2)	142	55	-3
Automated distribution hub (A_3)	184	70	-2